

```

test> use BookFlow
switched to db BookFlow
BookFlow> db
BookFlow
BookFlow> db.createCollection("book_metadata")
{ ok: 1 }
BookFlow> show collections
book_metadata
BookFlow> db.book_metadata.insertMany([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: "K. Rao",
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: "M1001",
        rating: 5,
        comment: "Excellent deep dive into digital systems design.",
        date: ISODate("2023-01-15")
      },
      {
        member_id: "M1042",
        rating: 4,
        comment: "Good but the Digital edition had formatting issues."
        date: ISODate("2023-02-20")
      }
    ]
  },
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: ISODate("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: ISODate("2022-06-01")
      }
    ]
  },
  {
    title: "Whispers of Yesterday",
    author: "S. Kapoor",
    genre: "Fiction",
    publishedYear: 2019,
    formats: ["Hardcover", "Paperback"],
    specs: {
      pages: 410,

```

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    specs: {
      pages: 410,
      binding: "Sewn"
    },
    avgRating: 3.8,
    reviews: [
      {
        member_id: "M3001",
        rating: 3,
        comment: "Slow start but strong ending.",
        date: ISODate("2020-01-15")
      },
      {
        member_id: "M3002",
        rating: 4,
        comment: "Beautifully written prose.",
        date: ISODate("2020-02-11")
      }
    ]
  },
  {
    title: "Quantum Horizons",
    author: "R. Menon",
    genre: "Science",
    publishedYear: 2023,
    formats: ["Digital", "Audiobook"],
    specs: {
      fileSize_MB: 5.6,
      drm: true,
      narrator: "P. Iyer",
      runtime_min: 540
    },
    avgRating: 4.9,
    reviews: [
      {
        member_id: "M4001",
        rating: 5,
        comment: "The Digital format made annotations so easy."
        date: ISODate("2023-08-01")
      },
      {
        member_id: "M4002",
        rating: 5,
        comment: "Mind-bending and well narrated.",
        date: ISODate("2023-09-14")
      }
    ]
  }
]
])
}
}
acknowledged: true,
insertedIds: {
  '0': ObjectId('6a95810136d38e4a76c72ab7'),
  '1': ObjectId('6a95810136d38e4a76c72ab8'),
  '2': ObjectId('6a95810136d38e4a76c72ab9'),
  '3': ObjectId('6a95810136d38e4a76c72aba')
}
}
BookFlow> db.book_metadata.find({ avgRating: { $gt: 4 } });
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
  },

```

```

BookFlow> db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } });
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72aba'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]

```

```

BookFlow> db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } });
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72aba'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]

```



```

BookFlow> db.book_metadata.find({ avgRating: { $gt: 4 }, formats: { $in: ["Digital"] } });
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72aba'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]

```

```
BookFlow> db.book_metadata.find({ formats: { $regex: "Digital", $options: "i" } });
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72aba'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]
```

```

BookFlow> db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } });
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72aba'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]

```

```
BookFlow> db.book_metadata.aggregate([{$match: { publishedYear: { $gt: 2020 } } }]);
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72aba'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]
```

```

BookFlow> db.book_metadata.aggregate([{$unwind: "$reviews" }]);
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: {
      member_id: 'M1042',
      rating: 4,
      comment: 'Good but the Digital edition had formatting issues.',
      date: ISODate('2023-02-20T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab9'),
    title: 'Whispers of Yesterday',
  }
]

```

```

BookFlow> db.book_metadata.aggregate([{$group: { _id: "$title", avgReviewRating: { $avg: "$reviews.rating" }, totalReviews: { $sum: 1 }, author: { $first: "$author" } } }]);
[
  {
    _id: 'Quantum Horizons',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'R. Menon'
  },
  {
    _id: 'Whispers of Yesterday',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'S. Kapoor'
  },
  {
    _id: 'The Silent Algorithm',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'R. Menon'
  },
  {
    _id: 'Ocean of Code',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'A. Fernandes'
  }
]
BookFlow> db.book_metadata.aggregate([{$sort: { avgReviewRating: -1 } }]);
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: ''
      }
    ]
  }
]

```

```

BookFlow> db.book_metadata.aggregate([{$project: { _id: 0, title: "$_id", author: 1, avgReviewRating: { $round: ["$avgReviewRating", 2] }, totalReviews: 1 } }]);
[
  {
    author: 'R. Menon',
    title: ObjectId('6a95810136d38e4a76c72ab7'),
    avgReviewRating: null
  },
  {
    author: 'A. Fernandes',
    title: ObjectId('6a95810136d38e4a76c72ab8'),
    avgReviewRating: null
  },
  {
    author: 'S. Kapoor',
    title: ObjectId('6a95810136d38e4a76c72ab9'),
    avgReviewRating: null
  },
  {
    author: 'R. Menon',
    title: ObjectId('6a95810136d38e4a76c72aba'),
    avgReviewRating: null
  }
]
BookFlow> db.book_metadata.aggregate([{$match: { publishedYear: { $gt: 2020 } } }, {$unwind: "$reviews" }, {$group: { _id: "$title", avgReviewRating: { $avg: "$review
s.rating" }, totalReviews: { $sum: 1 }, author: { $first: "$author" } } }, {$sort: { avgReviewRating: -1 } }, {$project: { _id: 0, title: "$_id", author: 1, avgReviewRa
ting: { $round: ["$avgReviewRating", 2] }, totalReviews: 1 } }]);
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,
    author: 'A. Fernandes',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  },
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  }
]
BookFlow> db.book_metadata.createIndex({ "reviews.comment": "text" });

```



```

BookFlow> db.book_metadata.createIndex({ "reviews.comment": "text" });
reviews.comment_text
BookFlow> db.book_metadata.find({ $text: { $search: "Digital" } });
[
  {
    _id: ObjectId('6a95810136d38e4a76c72ab7'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72aba'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a95810136d38e4a76c72ab8'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  }
]

```

```

BookFlow> db.book_metadata.find({ $text: { $search: "Digital" } }).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [ 'digit' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
      textIndexVersion: 3,
      inputStage: {
        stage: 'FETCH',
        nss: 'BookFlow.book_metadata',
        inputStage: {
          stage: 'IXSCAN',
          nss: 'BookFlow.book_metadata',
          keyPattern: { _fts: 'text', _ftsx: 1 },
          indexName: 'reviews.comment_text',
          isMultiKey: true,
          isUnique: false,
          isSparse: false,
          isPartial: false,
          indexVersion: 2,
          direction: 'backward',
          indexBounds: {}
        }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 133,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 1,
      restoreState: 1,
      isEOF: 1,
    }
  }
}

```

```

}
BookFlow> db.book_metadata.createIndex({ formats: 1 });
formats_1
BookFlow> db.book_metadata.createIndex({ avgRating: -1 });
avgRating_-1
BookFlow> db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 52,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'BookFlow.book_metadata',
      inputStage: {
        stage: 'IXSCAN',
        nss: 'BookFlow.book_metadata',
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultiKey: false,
        multiKeyPaths: { avgRating: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf, 4)' ] }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 54,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,
      stage: 'FETCH',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'BookFlow.book_metadata',
      docsExamined: 3,
      alreadyHasObj: 0,
      inputStage: {
        stage: 'IXSCAN',
        nReturned: 3,
        executionTimeMillisEstimate: 0,
        works: 4,
        advanced: 3,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        nss: 'BookFlow.book_metadata',
        keyPattern: { avgRating: -1 },

```

```

    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    keyPattern: { avgRating: -1 },
    indexName: 'avgRating_-1',
    isMultiKey: false,
    multiKeyPaths: { avgRating: [] },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: { avgRating: [ '[inf, 4)' ] },
    keysExamined: 3,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0,
    peakTrackedMemBytes: 0
  }
},
queryShapeHash: '79E1597FBF1E7E0551896CC4A5AA0604E417E5E37F937A039ABC2D303FB513AD',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'GNANESHWAR',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
BookFlow> db.book_metadata.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'reviews.comment_text',
    weights: { 'reviews.comment': 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  },
  { v: 2, key: { formats: 1 }, name: 'formats_1' },
  { v: 2, key: { avgRating: -1 }, name: 'avgRating_-1' }
]

```

```

BookFlow> db.book_metadata.aggregate([{$group: { _id: "$title", avgReviewRating: { $avg: "$reviews.rating" }, totalReviews: { $sum: 1 }, author: { $first: "$author" } } }]);
[
  {
    _id: 'Ocean of Code',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'A. Fernandes'
  },
  {
    _id: 'The Silent Algorithm',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'R. Menon'
  },
  {
    _id: 'Whispers of Yesterday',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'S. Kapoor'
  },
  {
    _id: 'Quantum Horizons',
    avgReviewRating: null,
    totalReviews: 1,
    author: 'R. Menon'
  }
]
BookFlow> { $avg: "$reviews.rating" }
$reviews.rating
BookFlow> db.book_metadata.aggregate([{$match: { publishedYear: { $gt: 2020 } } }, {$sumind: "$reviews" }, {$group: { _id: "$title", avgReviewRating: { $avg: "$reviews.rating" }, totalReviews: { $sum: 1 }, author: { $first: "$author" } } }, {$sort: { avgReviewRating: -1 } }, {$project: { _id: 0, title: "$_id", author: 1, avgReviewRating: { $round: ["$avgReviewRating", 2] }, totalReviews: 1 } }]);
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  },
  {
    totalReviews: 2,
    author: 'A. Fernandes',
    title: 'Ocean of Code',
    avgReviewRating: 4.5
  }
]
BookFlow> |

```